CSC 270-Artificial Intelligence

Building a Healthcare Knowledge Graph for AI - Cancer

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Phase 3: Schema and Ontology Design

Objective:

The objective of this phase is to design a comprehensive schema that defines the relationships between various healthcare entities, specifically focusing on lung cancer, its symptoms, treatments, medications, and associated disorders. A knowledge graph that can support improved comprehension, investigation, and clinical decision-making in oncology will be built around this schema.

Chosen Ontology:

For our knowledge graph, we will adopt a custom schema inspired by Schema.org's MedicalEntity vocabulary and extend it to suit the specific context of lung cancer. This schema will be designed to capture the intricate relationships and attributes relevant to diseases, symptoms, treatments, and medications.

Example Relationships:

→The following relationships will be established between entities:

1.Disease-HasSymptom → Symptom

- Description: This relationship indicates that a specific disease is associated with particular symptoms.
- Example: "Lung Cancer" → "Cough" indicates that coughing is a common symptom of lung cancer.

2.Disease-CanBeTreatedWith → Treatment

- Description: This relationship shows which treatments are applicable for a particular disease.
- Example: "Lung Cancer" → "Chemotherapy" signifies that chemotherapy is a treatment option for lung cancer.

3.Treatment-UsesMedication → Medication

- Description: This relationship connects treatments to the medications used within those treatments.
- Example: "Chemotherapy" → "Cisplatin" indicates that cisplatin is a medication used in chemotherapy for lung cancer.

4.Disease-Indicates → Associated Disorder

- Description: This relationship highlights disorders that can be a consequence or complication of the primary disease.
- Example: "Lung Cancer" → "Pulmonary Fibrosis" suggests that pulmonary fibrosis can be associated with lung cancer.

5.Treatment-CanBeCombinedWith → Treatment

- Description: This relationship illustrates treatments that can be administered in combination.
- Example: "Immunotherapy" → "Chemotherapy" shows that immunotherapy can be combined with chemotherapy in treating lung cancer.

Entity Attributes

- To ensure a robust representation of each entity type, the following attributes will be defined:

1. **Disease**

- Name: The official name of the disease (e.g., "Lung Cancer").
- Severity: A classification of the disease's severity, which could be detailed into stages (e.g., "Stage I", "Stage II", "Advanced").
- Common Symptoms: A list of symptoms typically associated with the disease (e.g., "Cough", "Shortness of breath", "Chest pain").
- Associated Disorders: Disorders that frequently occur alongside the disease (e.g., "Pulmonary Fibrosis", "Metastatic Lung Cancer").

2. **Symptom**

- Description: A detailed explanation of the symptom's nature (e.g., "Coughing that produces phlegm").
- Severity: An assessment of how severe the symptom is (e.g., "Mild", "Moderate", "Severe").
- Duration: The typical duration that the symptom lasts (e.g., "Acute" for short-term symptoms, "Chronic" for long-lasting symptoms).

3. **Treatment**

- Name: The name of the treatment modality (e.g., "Chemotherapy", "Immunotherapy").
- Effectiveness: Quantitative or qualitative measures of how effective the treatment is for the disease (e.g., "70% response rate").
- Side Effects: A comprehensive list of potential side effects associated with the treatment (e.g., "Nausea", "Fatigue", "Hair loss").

4. **Medication**

- Name: The name of the medication (e.g., "Cisplatin", "Atezolizumab").
- Dosage: Recommended dosage information (e.g., "100 mg/m² every 3 weeks").
- Drug Interactions: Known interactions with other medications that patients may be taking (e.g., "Cisplatin should not be used with nephrotoxic agents").

→This structured schema will facilitate the development of a knowledge graph capable of supporting clinical research, patient education, and data analysis in the field of oncology, particularly concerning lung cancer.